

```
!nvidia-smi
```

Thu Apr 9 09:29:51 2026

NVIDIA-SMI 580.82.07			Driver Version: 580.82.07			CUDA Version: 13.0		
GPU	Name		Persistence-M	Bus-Id	Disp.A	Volatile	Uncorr.	ECC
Fan	Temp	Perf	Pwr:Usage/Cap		Memory-Usage	GPU-Util	Compute M.	MIG M.
=====								
0	Tesla T4		Off	00000000:00:04.0	Off			0
N/A	38C	P8	9W / 70W		0MiB / 15360MiB	0%	Default	N/A
=====								
Processes:								
GPU	GI	CI	PID	Type	Process name	GPU Memory		
	ID	ID				Usage		
=====								
No running processes found								
=====								

```
from numba import cuda
import numpy as np
```

```
import torch

print(torch.cuda.get_device_name(0))
```

Tesla T4

```
import torch

props = torch.cuda.get_device_properties(0)
print(props.max_threads_per_block)
```

1024

```
import tensorflow as tf

print(tf.config.list_physical_devices('GPU'))

[PhysicalDevice(name='/physical_device:GPU:0', device_type='GPU')]
```

```
import torch

props = torch.cuda.get_device_properties(0)

print("GPU:", props.name)
print("Max threads/block:", props.max_threads_per_block)
print("Multiprocessors:", props.multi_processor_count)
```

GPU: Tesla T4
Max threads/block: 1024
Multiprocessors: 40

```
from numba import cuda
import numpy as np

# CUDA kernel
@cuda.jit
def reduce_sum(arr):
    tid = cuda.threadIdx.x

    # Parallel reduction
    stride = 256
    while stride > 0:
        if tid < stride:
            arr[tid] += arr[tid + stride]
            cuda.syncthreads()
            stride //= 2

# Host data: numbers 1 to 512
data = np.arange(1, 513, dtype=np.int32)
```

```
# Sum on GPU
```

```
# Copy to GPU
d_data = cuda.to_device(data)

# Launch kernel: 1 block, 512 threads
reduce_sum[1, 512](d_data)

# Copy back result
result = d_data.copy_to_host()

print("Sum =", result[0])
```

```
/usr/local/lib/python3.12/dist-packages/numba_cuda/numba/cuda/dispatcher.py:696: NumbaPerformanceWarning: Grid s
    warn(errors.NumbaPerformanceWarning(msg))
Sum = 131328
```